AIM:

To create a dimensioned sketch of the given Bracket 2 profile using Autodesk Fusion 360.

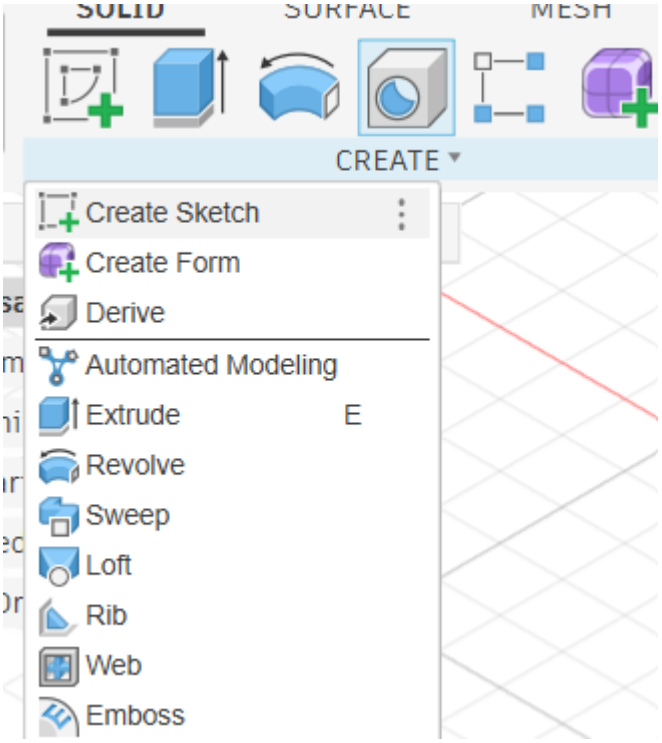
REQUIREMENTS:

1. Autodesk Fusion 360 software.
2. Windows 10 operating system.
3. 16 GB RAM with suitable graphics support.
4. Stable internet connection.

PROCEDURE:

STEP 1: Create a New Sketch

1. Open Autodesk Fusion 360 and create a new design.
2. Select Create Sketch and choose the XY plane.
3. Draw horizontal and vertical construction centre lines through the origin.
4. Use these centre lines as references for placing the circular holes and central profile.

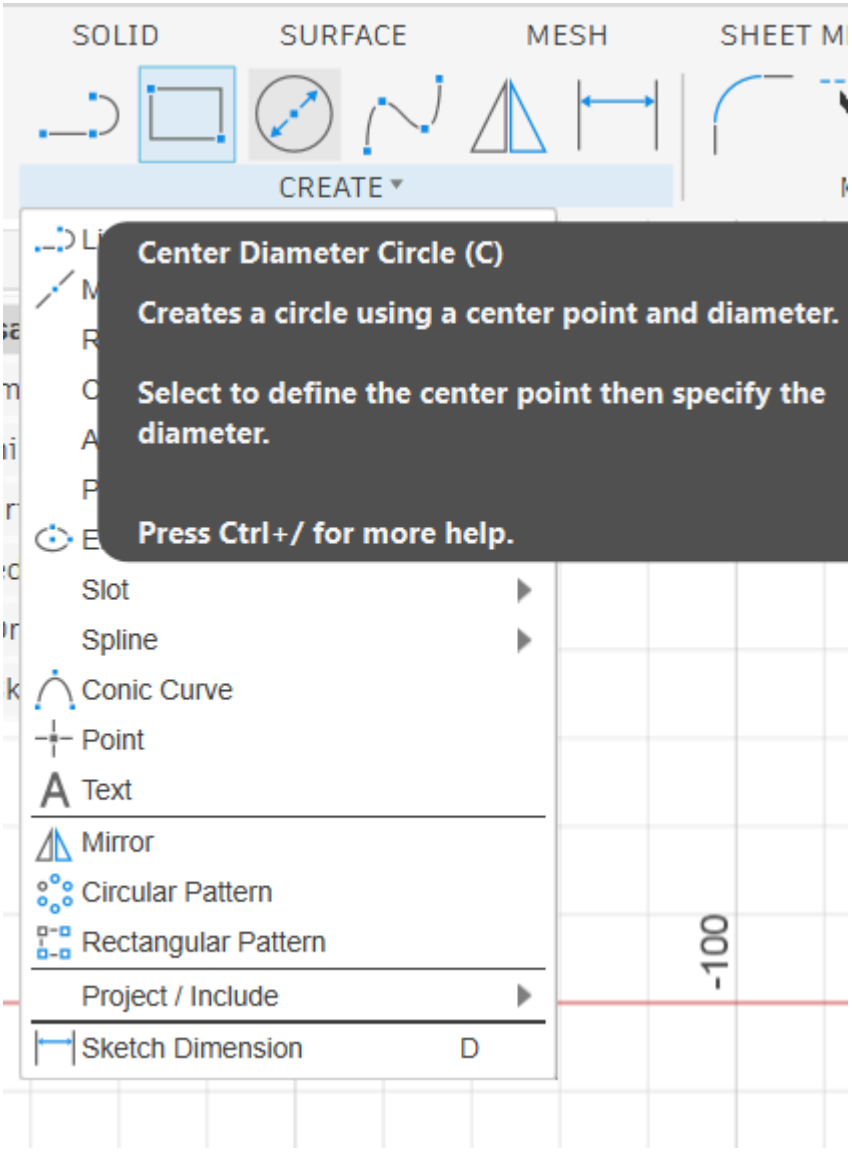


STEP 2: Construct the Central Circular Profile

1. Select the Circle tool and draw the central circle at the origin.
2. Add the surrounding circular profile using the radius dimensions indicated in the given drawing.
3. Draw one curved slot around the central circle using the Arc tool.
4. Use the Offset and Trim tools wherever necessary to create the required slot profile.
5. Apply the Circular Pattern tool to create six equally spaced curved slots around the centre.

STEP 3: Create the Six Outer Circular Profiles

1. Use the Circle tool to draw six outer circular profiles in the positions shown in the drawing.
2. Arrange two circles at the top, two at the middle and two at the bottom.
3. Draw the inner holes concentrically inside the outer circles.
4. Apply the specified dimensions, including R15 and R21, according to the drawing.
5. Add centre marks and construction lines to maintain symmetry.



STEP 4: Draw the Upper Connecting Curves

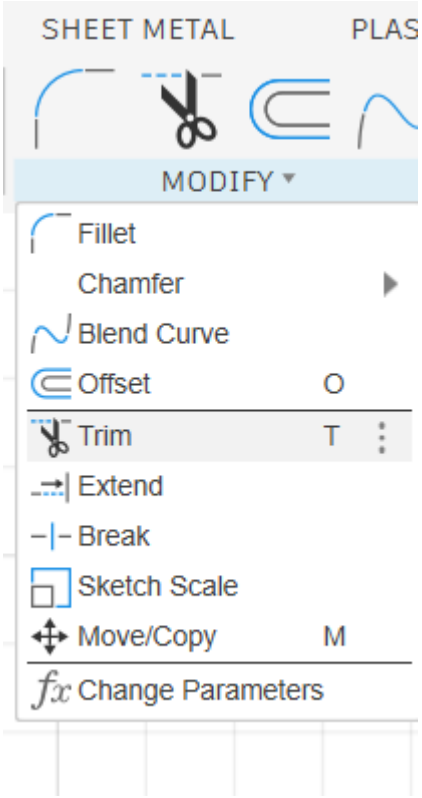
1. Select the Arc tool to connect the two upper circular profiles.
2. Create the curved depression between the upper circles using the specified R20 radius.
3. Connect the upper circles to the middle circular profiles using smooth outer arcs.
4. Apply tangent constraints to obtain smooth transitions between the connected profiles.

STEP 5: Draw the Middle and Lower Connecting Curves

1. Draw the connecting arcs between the middle circular profiles and the lower circular profiles.
2. Use the required R20 and R21 dimensions wherever indicated.
3. Create the two lower curved portions using the R20 radius shown in the drawing.
4. Ensure that all arcs connect smoothly with the circular profiles.
5. Apply the required dimensions and constraints to maintain the shape and symmetry.

STEP 6: Complete the Outer Boundary

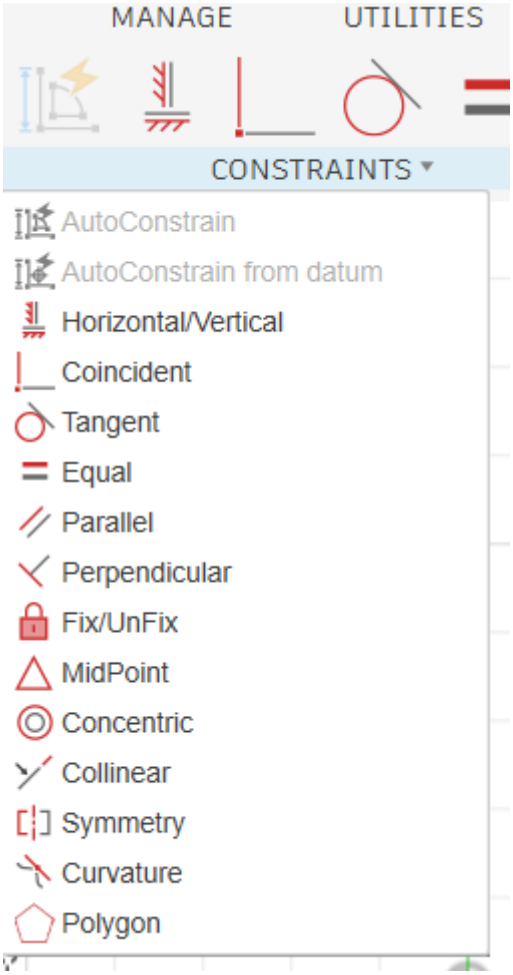
1. Use the Line and Arc tools to complete the outer bracket profile.
2. Connect the upper, middle and lower portions according to the given drawing.
3. Apply the specified radius dimensions, including R40, wherever indicated.
4. Use the Trim tool to remove unwanted overlapping lines and arcs.
5. Check that the outer boundary is continuous and all six holes are correctly positioned.



STEP 7: Apply Dimensions and Constraints

1. Select Sketch Dimension (D) to apply the required dimensions.
2. Add the radius dimensions R15, R16, R20, R21, R32 and R40 as indicated in the drawing.
3. Apply coincident, tangent and symmetry constraints wherever required.

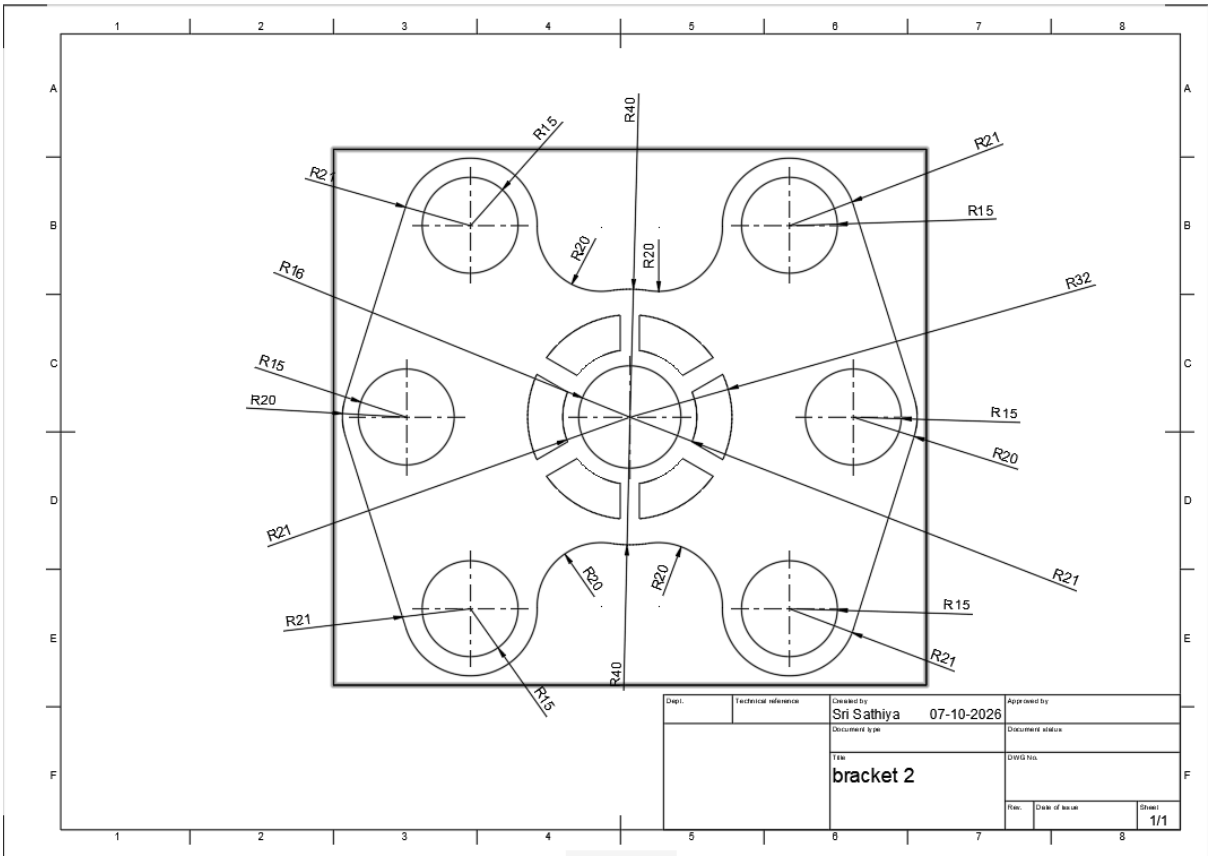
4. Verify the central slots, circular holes and outer profile against the reference drawing.



STEP 8: Finish and Create the Drawing

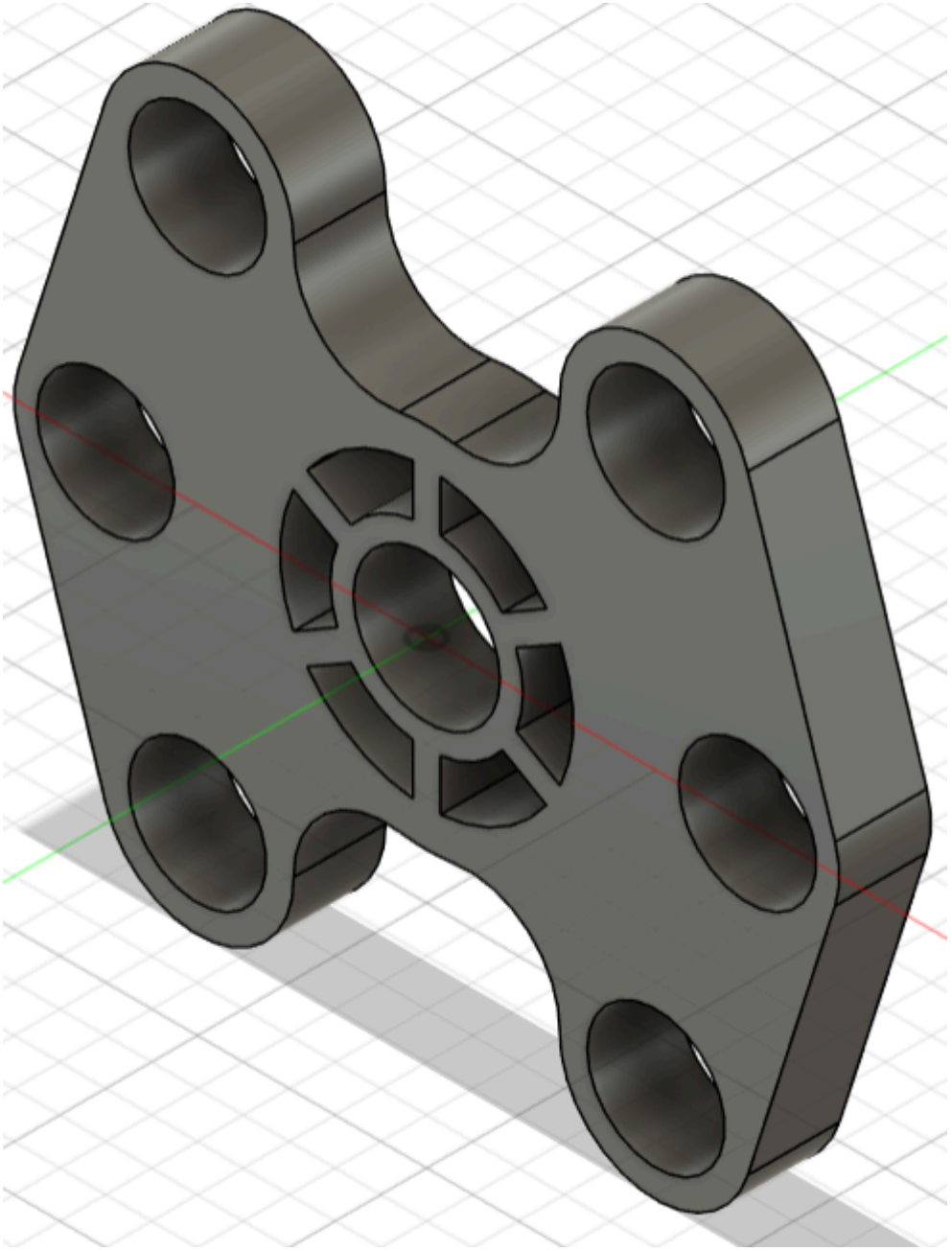
1. Check the completed sketch and click Finish Sketch.
2. Open the Drawing workspace and select From Design.
3. Choose the required ISO drawing standard and place the drawing view on the sheet.

4. Add the necessary dimensions and save the completed drawing.



OUTPUT:

The dimensioned Bracket 2 profile was created using Autodesk Fusion 360.



RESULT:

Thus, the given Bracket 2 profile was sketched, dimensioned and drafted using Autodesk Fusion 360.